PROJECT ATLAS

ATLAS-DES-003: TIMING LOGIC SPECIFICATION

1. Timing Philosophy

In accordance with **UR-DT-03**, the timing engine must be deterministic. The system shall not rely on simple software sleep loops. Instead, it shall utilize a **Reference Clock Synchronization** method.

2. The Pulse (QTimer Implementation)

The `AtlasTimerEngine` will run at 10Hz (every 100ms). While the display only updates once per second, the 10Hz "pulse" ensures that when the PC clock moves to the next second, Atlas reacts within 0.1s, maintaining the illusion of perfect synchronization.

Mathematical Definition: The Countdown

```
Time_Remaining = Target_End_Time - Current_PC_Time
```

By calculating the remaining time on every pulse (rather than just subtracting 1 from a counter), the system is immune to CPU-induced lag or drift.

3. State-Timer Interaction

State	Timer Action	Display Output
READY / ARMED	Static	Target Insertion Window (e.g. 15:00)
RUNNING (Window)	Active Countdown	Insertion Window T-Minus
RUNNING (Heat)	Active Countdown	Heat Duration T-Minus
CANCELLED	Zeroed	00:00:00

4. Precision Requirements

- Operating frequency: 10Hz.
- Allowable drift over 60 minutes: < 100ms.
- Sync point: Python `time.time()` or `QDateTime.currentDateTime()`.